

Chapter 14: Human Communication

Learning Objective

Detailed Answer Key

14.1 Compare the language-related functions of the left and right hemispheres, and discuss the prevalence of these functions in the opposite hemisphere.

Left Hemisphere Functions: The left hemisphere typically controls language comprehension, the analysis of speech, and language production. **Right Hemisphere Functions:** The right hemisphere controls activities like reading maps, perceiving spatial relations, recognizing complex geometrical forms, organizing a narrative, and understanding **prosody**—the expression and recognition of emotion in the tone of voice. **Prevalence of Lateralization:** In approximately **90 percent** of the population, the left hemisphere is dominant for speech and language functions. Right hemisphere speech dominance is rare, found in only **4 percent** of right-handed people, 15 percent of ambidextrous people, and **27 percent** of left-handed people.

14.2 List factors involved in language production, and identify corresponding brain regions.

Language production requires an individual to have something to communicate, which can be a current event, a past event (memories), or an imagined event. Perceptions of current and past events rely on the **occipital, temporal, and parietal lobes** of the brain. Although little is known about the exact mechanisms of imagination, it likely employs the same brain areas as perceptions and memories. The conversion of these perceptions, memories, and thoughts into language involves specific brain regions.

14.3 Describe the factors involved in language comprehension, and identify corresponding brain regions.

Language comprehension requires **recognizing words** (which can be auditory or visual) and **understanding their meanings**. Word recognition utilizes **Wernicke's area** in the left temporal lobe. The meanings associated with words are generally stored as memories in the sensory association areas. Comprehension of the **figurative aspects of language** (like metaphors, proverbs, and moral

stories) involves the **right hemisphere**. Studies confirm that evaluating metaphors activates the right superior temporal cortex, and transcranial magnetic stimulation (TMS) applied here impairs the understanding of novel metaphors.

14.4 Describe evidence for common and language-specific brain regions for bilingual language processing.

The majority of people are bilingual or multilingual, and imaging studies reveal that some brain regions used for language processing are **devoted to specific languages**, while others are **common to both**. Both common and language-specific areas are found in the **frontal and posterior temporal/parietal cortexes**, with some language-specific areas also found in subcortical structures. Direct cortical electrostimulation performed during brain surgery has confirmed the existence of both common and language-specific naming sites in bilingual patients. Additionally, studies show that learning a second language is associated with increased gray matter density in the **left parietal cortex**, particularly among those who learned early in life and are highly proficient.

14.5 Identify brain structures and functions involved in prosody.

Prosody refers to the **rhythmic, emphatic, and melodic aspects of speech**. It is primarily a function of the **right hemisphere**. This function is likely related to musical skills and the expression and recognition of emotions. Functional MRI studies show that the prosodic components of speech primarily activate the **right hemisphere** (orange and yellow regions in the scans shown in the source material), while meaningful components activate the left hemisphere (blue and green regions). Interestingly, damage to the left hemisphere (such as in Broca's aphasia) can also result in deficits in prosody.

14.6 Identify the brain regions involved in recognizing people's voices.

The perception of a person's identity from their voice is independent of word recognition. Regions of the **right hemisphere** are involved in recognizing specific voices. Damage to these areas causes **phonagnosia**. Functional-imaging studies specifically implicate the **right anterior superior temporal cortex** in voice recognition.

14.7 Describe the symptoms and neural mechanisms of Broca's aphasia, including agrammatism, anomia, and articulation.

Broca's aphasia is a disturbance in language production characterized by **slow, laborious, and nonfluent speech**. Patients typically **comprehend language better than they produce it**. It is caused by damage to the **frontal lobe** (specifically Broca's area) and/or underlying subcortical white matter and basal ganglia involvement. The symptoms include: 1. **Agrammatism:** Difficulty using grammatical devices (function words like articles, pronouns, and verb endings), resulting in a reliance on **content words** (nouns, main verbs). 2. **Anomia:** Difficulty finding or recalling the appropriate word (word-finding difficulty); patients often use **circumlocution** (roundabout speaking). 3. **Articulation Difficulties:** Problems with the **motor memories** for sequences of muscular movements needed to articulate words.

14.8 Describe the symptoms and neural bases of Wernicke's aphasia, including deficits in spoken word recognition, comprehension, and converting thoughts into words.

Wernicke's aphasia (receptive aphasia) is characterized by **poor language comprehension** and the production of fluent but **meaningless speech**. It results from damage to **Wernicke's area** in the left hemisphere and the **posterior language area**. Afflicted individuals are often unaware of their deficits. The disorder involves three key deficits: 1. **Deficits in Spoken Word Recognition:** Damage limited to Wernicke's area or disruption of auditory input results in **pure word deafness**, where a person can perceive speech sounds but cannot recognize words, though their own speech, reading, and writing are unaffected. 2. **Deficits in Comprehension of**

Meaning: Damage extending to the **posterior language area** (junction of temporal, occipital, and parietal lobes) disrupts the ability to interchange information between the auditory representation of words (Wernicke's area) and their meanings (stored memories in the sensory association cortex). Damage limited here results in **transcortical sensory aphasia**, where the person can repeat words but cannot understand them or produce meaningful speech. 3. **Deficits in Converting Thoughts to Words:** The inability to connect thoughts/memories (stored in the sensory and motor association cortexes) to their corresponding words for meaningful speech production.

Symptoms: Conduction aphasia features **fluent and meaningful speech** with **good comprehension**, but **poor repetition** (especially of nonwords). **Neural**

Basis: It is caused by damage to the **arcuate fasciculus**, the fiber bundle connecting Wernicke's area and Broca's area. The symptoms suggest that there are **two pathways** between these language areas: 1. **Direct Pathway (Arcuate Fasciculus):** Conveys information about the **sounds of words** from Wernicke's area to Broca's area for repetition (a **phonological loop**). Damage to this pathway impairs repetition (conduction aphasia). 2. **Indirect Pathway (via Posterior Language Area):** This route is based on the **meaning of words** and bypasses the damaged arcuate fasciculus, allowing the patient to comprehend words and express thoughts using alternative word choices (semantic paraphasias).

14.9 Identify the symptoms of conduction aphasia, and describe their contribution to understanding about connections between Broca's and Wernicke's areas.

14.10 Explain the brain processes involved in production and comprehension of American Sign Language.

ASL is expressed by **hand movements** and functions as a full language. The **left hemisphere** plays the dominant role in both the production and comprehension of sign language, similar to spoken language. **Mirror neurons** found in the **ventral**

premotor cortex (including Broca's area) are activated when a person observes or imitates finger movements, suggesting they are involved in acquiring sign language. The **right hemisphere** contributes to the figurative and narrative aspects of ASL, similar to its role in spoken language prosody.

14.11 Describe the biological basis for stuttering and identify treatment strategies.

Biological Basis: Stuttering is influenced by **genetic factors** and affects about 1% of the population. It is characterized by pauses and repetitions that disrupt fluent speech. The problem involves **abnormalities in neural mechanisms involved in planning and initiating speech**. Functional imaging shows excessive activation in the inferior frontal cortex (**Broca's area**) and other motor regions, but **decreased activation in the auditory regions of the temporal lobe**. This suggests a problem with **faulty auditory feedback** from the person's own speech. **Treatment:** Treatments focus on improving fluency. **Fluency-shaping therapy** successfully leads to increased activation in the superior temporal lobe (auditory regions) that previously showed low activity. Devices that provide **delayed auditory feedback** can also facilitate speech fluency in some people who stutter.

14.12 Compare aphasia and disorders of reading and writing.

Aphasias (disorders of comprehension or production of language) are usually accompanied by **parallel deficits in reading and writing**. For instance, Wernicke's aphasia patients have difficulty comprehending language whether spoken or written. Similarly, if a person's speech (Broca's aphasia) is agrammatical, so is their writing. However, distinct disorders exist: **Pure alexia** (cannot read, can write) and **pure word deafness** (cannot understand spoken words, can read and write) demonstrate that these faculties involve separate brain mechanisms.

14.13 Identify the symptoms of pure alexia, and describe the brain structures involved.

Symptoms: Pure alexia is the **loss of the ability to read** without the loss of the ability to write. Patients with pure alexia can often still recognize words spelled aloud. It is a perceptual disorder where visual information is prevented from reaching the visual association cortex. **Brain Structures:** This deficit is caused by damage to the **left primary visual cortex** (causing blindness in the right visual field) and the **posterior corpus callosum** (which normally relays visual information from the right visual cortex to the left hemisphere). The core problem is preventing visual input from reaching the **visual word-form area (VWFA)** in the left hemisphere.

14.14 Describe how research on acquired and developmental dyslexia helps us to understand the role of the brain in reading.

Research on **acquired dyslexias** (caused by brain damage in already literate individuals) and **developmental dyslexias** (present early in life, often with a genetic origin) reveals the dual processing nature of reading: **whole-word reading** and **phonetic reading** (sounding out letters). - **Whole-Word Reading (Lexical Route):** This recognizes familiar words as a whole and involves the **Visual Word-Form Area (VWFA)** in the fusiform gyrus. Damage here causes **surface dyslexia** (difficulty reading irregularly spelled words like "sew," but intact phonetic reading). - **Phonetic Reading (Phonological Route):** This involves sounding out unfamiliar words. Damage here causes **phonological dyslexia** (intact whole-word reading but inability to sound out words or nonwords). **VWFA Plasticity:** Reading relies on neural plasticity, where the VWFA, originally evolved for object recognition, is "recycled" to recognize the configurations of lines in written characters (like letters or Kanji). Developmental dyslexia is associated with decreased activation in the **left temporoparietal cortex** and the **VWFA**.

14.15 Explain how research on audition, vision, memorization, and motor control helps us to understand the role of the brain in writing.

Writing involves coordinating **audition, vision, memorization, and motor control**. **Vision/Motor Control:** Simply viewing alphabetical characters activates the **ventral premotor cortex**, suggesting that visual and motor programs are highly integrated, particularly in the hemisphere controlling the dominant hand. Deficits include: - **Phonological Dysgraphia (Audition/Phonetic):** Inability to sound out words phonetically, typically involving damage to Broca's area and the insula. - **Orthographic Dysgraphia (Vision/Whole-Word):** Difficulty with visual-based writing (irregularly spelled words like *busy*), implicating the **posterior inferior temporal cortex** and the VWFA. **Memorization:** The memory for typing (dystypia) or texting (dystextia) certain words seems linked to specific motor memories located in the left frontal/parietal lobe region.

Chapter 15: Neurological Disorders

Learning Objective

Detailed Answer Key

15.1 Describe the primary symptoms, causes, and treatments for brain tumors.

Symptoms/Damage: Tumors damage brain tissue through **compression** (pushing against the brain, causing indirect damage or blocking CSF flow) or **infiltration** (invading and destroying tissue). **Causes:** Tumors do not originate from neurons, but from **glial cells** (gliomas, the most serious) or **metastases** from other parts of the body. They are classified as **benign** (encapsulated, easily removed) or **malignant** (infiltrating, difficult to remove, can metastasize). **Treatments:** Treatment involves **surgical removal** (especially for benign tumors) and targeting remaining cells with **radiation**. Some tumors respond to chemotherapy.

15.2 Describe the primary symptoms, causes, and treatments for seizures.

Symptoms: A seizure is a period of **sudden, excessive neuronal activity**. A severe seizure may cause a **convulsion** (uncontrollable muscular movement). Seizures are broadly categorized as **partial** (starting from a

definite focus, potentially complex leading to loss of consciousness, or simple partial with changes in consciousness) or **generalized** (widespread involvement, like a Grand Mal seizure with tonic and clonic phases).

Causes: The most common cause is **scarring** (from injury, stroke, or tumors). Genetic factors influencing ion channels contribute to incidence, but most cases are non-genetic. **Treatments:** Treatment involves **anticonvulsant drugs** that increase the effectiveness of inhibitory synapses (often GABAergic synapses). **Brain surgery** to remove the seizure focus or **vagus nerve stimulation (VNS)** can also reduce seizure activity.

15.3 Explain how cerebrovascular accidents can occur.

Cerebrovascular accidents (strokes) are caused by vascular problems and are primarily categorized as **hemorrhagic** or **ischemic**. 1. **Hemorrhagic Strokes:** Caused by **bleeding within the brain** due to a malformed or weakened blood vessel (often due to high blood pressure), where accumulating blood puts pressure on and damages surrounding tissue. 2. **Ischemic Strokes:** Caused by **occlusion (blockage) of a blood vessel**. This can be due to a **thrombus** (a clot forming locally in a vessel) or an **embolus** (a piece of material that breaks off elsewhere and travels to a narrower artery). Neuronal death is immediately caused by **excitotoxicity** (excessive glutamate release) resulting from blood loss.

15.4 Explain how treatments can be used to address the immediate and long-term symptoms of cerebrovascular events.

Immediate Treatments (Acute Phase): Focus on restoring blood flow and minimizing damage. For ischemic strokes, immediate intervention may include injecting a clot-dissolving drug like **tPA** (tissue plasminogen activator) within 3–4.5 hours, or removing the occlusion using medical devices (coils, aspiration, stents). **Long-Term Treatments (Recovery/Rehabilitation):** Includes drugs to reduce swelling/inflammation. **Physical, speech, and/or occupational therapy** is crucial, such as **constraint-induced movement therapy** (forcing use of the affected limb to "rewire" the brain). Additionally, **brain-computer**

interfaces are being developed to help patients control prosthetic devices.

15.5 Identify some causes of traumatic brain injury.

TBI can result from **open-head injuries** (projectile/fractured skull wounding the brain) or **closed-head injuries** (blunt trauma causing the brain to strike the skull, known as coup and contrecoup injuries). **Chronic Traumatic Encephalopathy (CTE)** is a form of TBI caused by **repeated head trauma** (often from contact sports or blast injuries). CTE involves neurodegeneration characterized by the accumulation of **abnormal tau protein** in the cortex and limbic system.

15.6 Describe treatments for traumatic brain injuries.

Primary Treatments (Immediate): Focus on reducing **swelling** and **intracranial pressure** and ensuring **adequate blood flow**. **Secondary Treatments (Long-Term):** Address symptoms arising after injury. Drugs that inhibit **glutamate release** can prevent excitotoxicity, and the long-term cognitive and behavioral deficits are addressed using **rehabilitation strategies** similar to those for stroke (e.g., behavioral intervention and therapy).

15.7 Describe the effects of alcohol on development of the nervous system.

Alcohol is a **teratogen** whose presence during pregnancy can **impair fetal development** and cause **intellectual disability**. Consumption of alcohol, sometimes even a single binge, during a critical period of fetal development can lead to **fetal alcohol spectrum disorder (FASD)**. This causes abnormal facial development and impaired brain development.

15.8 Contrast the symptoms, causes, and treatments of inherited metabolic disorders.

These disorders are caused by inherited mutations leading to the **absence of an enzyme** necessary for constructing or breaking down specific chemical compounds.

Phenylketonuria (PKU): Caused by the absence of an enzyme converting phenylalanine to tyrosine, leading to accumulation of phenylalanine that interferes with CNS myelination. The treatment is an **early diagnosis** and a strict **low-phenylalanine diet**. **Tay-Sachs disease:** Caused by missing enzymes that lead to the accumulation

of waste substances in the brain. The brain swells and becomes damaged, eventually leading to death. (See summary Table 15.3).

Cause: Down syndrome is a congenital disorder caused by the presence of an **extra twenty-first chromosome**.

Symptoms: It results in abnormal development of the brain, leading to varying degrees of **intellectual disability**.

15.9 Identify the cause, symptoms, and interventions for Down syndrome.

After age 30, the brain begins to degenerate, resembling the pathology of **Alzheimer's disease**. **Interventions:**

Some drug treatments have shown promise, such as **GABA antagonists** (to enhance synaptic plasticity) or the **SSRI fluoxetine** (to reduce GABA concentrations), aiming to improve learning and memory, though GABA antagonists pose a seizure risk.

15.10 Describe how transmissible spongiform encephalopathies cause brain damage.

TSEs are contagious, fatal brain diseases caused by **prions** (protein infectious agents). Prions are normal proteins that become **misfolded** (PrP^{Sc}). Once misfolded, the infectious prion protein causes other normal proteins (PrP^C) to misfold as well, which ultimately **kills the neurons**. The accumulated misfolded protein triggers **apoptosis** (cell suicide).

15.11 Describe the symptoms, causes, and treatments for Parkinson's disease.

Symptoms: Muscular rigidity, **slowness of movement (akinesia)**, a **resting tremor**, and postural instability. It results from degeneration of the **nigrostriatal dopaminergic system** (neurons from the substantia nigra projecting to the basal ganglia). Surviving neurons show abnormal, circular **Lewy bodies**. **Causes:** Most cases are sporadic, but hereditary forms exist, notably mutations in: 1. **α -synuclein** (toxic gain of function; protein misfolds and aggregates). 2. **Parkin** (loss of function; failure to correctly tag defective proteins for destruction by proteasomes leads to protein accumulation). **Treatments:** 1. **L-DOPA:** Dopamine precursor, alleviating symptoms by increasing dopamine release from surviving neurons. 2. **Surgical Procedures:** Transplantation of fetal tissue; lesioning inhibitory nuclei (e.g., internal globus pallidus);

or **Deep Brain Stimulation (DBS)** of the subthalamic nucleus or globus pallidus.

15.12 Describe the symptoms, causes, and treatments for Huntington's disease.

Symptoms: Progressive **uncontrollable jerking movements**, writhing, cognitive changes, and eventual death. Degeneration begins in the **putamen**, particularly affecting GABAergic medium spiny neurons. **Causes:** A hereditary disorder caused by a **dominant gene on chromosome 4**. This mutation leads to repeated CAG sequences, producing abnormal huntingtin (Htt) protein, which results in a **toxic gain of function** and triggers apoptosis. **Treatments:** There is currently no cure. Experimental treatments involve **Happ1 intrabodies** or injection of **small interfering RNAs (siRNA)** or **antisense oligonucleotides** to block the production of the mutant Htt protein.

15.13 Describe the symptoms, causes, and treatments for amyotrophic lateral sclerosis.

Symptoms: ALS attacks spinal cord and cranial nerve motor neurons, leading to **spasticity, progressive weakness, muscular atrophy, and eventual paralysis** (respiratory failure is often the cause of death). **Causes:** Most cases are sporadic (90%), but some are hereditary, involving a mutation of the gene that produces **superoxide dismutase 1 (SOD1)**, leading to a toxic gain of function. The disease damages cells primarily through **glutamate excitotoxicity**. **Treatments:** The only current pharmacological treatment is **riluzole**, a drug that reduces glutamate-induced excitotoxicity by decreasing glutamate release, slightly prolonging life expectancy.

15.14 Describe the symptoms, causes, and treatments for multiple sclerosis.

Symptoms: MS is an **autoimmune demyelinating disease** where the immune system attacks myelin sheaths in the central nervous system (CNS), leaving **sclerotic plaques**. Symptoms vary widely and often follow a **remitting-relapsing pattern**. **Causes:** An autoimmune reaction, possibly triggered by a **virus** contracted in childhood. The damage is confined to CNS myelin. **Treatments:** Treatments are only partially effective and include **interferon β** (modulates the immune

system) and **glatiramer acetate** (suppresses immune cells that attack myelin).

15.15 Describe the symptoms, causes, and treatments for Alzheimer's disease.

Symptoms: Progressive **loss of memory** (especially recent events) and other cognitive functions, eventually leading to death. Characterized by **amyloid plaques** (β -amyloid protein) and **neurofibrillary tangles** (tau protein). **Causes:** Associated with genetic mutations in the gene for **APP** (on chromosome 21) or genes for **presenilins**. The presence of the **APOE4 allele** increases risk, possibly causing an imbalance leading to excessive **long-form β -amyloid**. Chronic TBI, obesity, hypertension, and low education are risk factors.

Treatments: Current medications (e.g., acetylcholinesterase inhibitors like donepezil and NMDA receptor antagonists like memantine) provide **slight symptomatic improvement** but do not halt neural degeneration. Immunological research pursuing **vaccines or antibody drugs** (e.g., aducanumab) aimed at destroying β -amyloid protein is ongoing, but initial vaccine trials led to inflammatory reactions.

15.16 Describe the symptoms and causes of Korsakoff's syndrome.

Symptoms: **Permanent anterograde amnesia** (inability to form new declarative memories), often accompanied by retrograde amnesia. **Causes:** Caused by a **thiamine (vitamin B₁) deficiency**, typically due to chronic alcohol consumption and/or malnutrition. The characteristic brain damage is seen in the **mammillary bodies**.

15.17 Identify the symptoms, causes, and treatments for encephalitis.

Symptoms: Encephalitis is an **infection/inflammation of the entire brain**. Symptoms include fever, irritability, nausea, often followed by convulsions, delirium, and signs of brain damage (e.g., aphasia or paralysis). **Causes:** Caused by **viruses** (like Herpes simplex, polio, or rabies) or toxic chemicals. **Treatments:** There is no specific treatment; only **supportive care** is provided.

15.18 Identify the symptoms, causes, and treatments for meningitis.

Symptoms: Meningitis is **inflammation/swelling of the meninges** (the connective tissue surrounding the CNS). Symptoms include a severe **headache, a stiff neck** (due to stretching inflamed meninges), and sometimes convulsions or confusion. **Causes:** Can be caused by **viruses or bacteria**; bacterial meningitis often causes brain damage. **Treatments:** Treatment involves antibiotics (for bacterial origin) or supportive care for viral meningitis.

Chapter 16: Schizophrenia, Affective Disorders, Anxiety Disorders, and OCD

Learning Objective

Detailed Answer Key

16.1 Describe the positive, negative, and cognitive symptoms of schizophrenia.

Schizophrenia symptoms fall into three categories: 1. **Positive Symptoms:** Behaviors present in excess of typical functioning. These include **thought disorders** (disorganized, irrational thinking), **delusions** (false beliefs like persecution or control), and **hallucinations** (perception of nonexistent stimuli). 2. **Negative Symptoms:** The absence or decrease of typical behaviors. These include social withdrawal, **lack of affect** (reduced emotional response), **poverty of speech**, and reduced motivation. 3. **Cognitive Symptoms:** Deficits in executive function. These include difficulty sustaining **attention**, low **psychomotor speed**, deficits in **learning and memory**, and poor abstract thinking or problem-solving.

16.2 Describe genetic factors associated with the development of schizophrenia.

Schizophrenia is highly heritable, though no single "schizophrenia gene" exists. **Twin Studies** show a genetic component: rates of schizophrenia are significantly higher in **monozygotic twins** (sharing a single placenta, 60% concordance) compared to **dichorionic monozygotic twins** (10.7% concordance), indicating a prenatal environmental interaction. **Mutations** in genes like **DISC1** (Disrupted in Schizophrenia 1) increase the likelihood of the disorder. **Paternal age** is also a factor, as children of older fathers are more likely to develop

schizophrenia due to increased likelihood of mutations in sperm cells.

16.3 Describe environmental factors associated with the development of schizophrenia.

Environmental factors that increase the risk of schizophrenia include the **seasonality effect** (increased incidence in people born during late winter/early spring), **viral epidemics** (e.g., after influenza outbreaks), and **vitamin D deficiency** (which plays a role in brain development). Other factors include living in high **population density** (large cities), **prenatal stress or malnutrition**, and maternal **tobacco use** during pregnancy.

16.4 Provide evidence of the role of the mesolimbic dopamine pathway in the positive symptoms of schizophrenia.

The **Dopamine Hypothesis** posits that positive symptoms (delusions, hallucinations) are caused by the **hyperactivity of dopaminergic synapses in the mesolimbic pathway** (VTA $\rightarrow$ nucleus accumbens and amygdala). Evidence includes: 1. **Pharmacology:** Dopamine **agonists** (like amphetamine and cocaine) produce/reinforce positive symptoms, while dopamine **antagonists** (like chlorpromazine and other antipsychotics) reduce them by blocking D2 and D3 receptors. 2. **Dopamine Release:** Patients with schizophrenia show **elevated dopamine release** in the striatum (including nucleus accumbens) when given amphetamine, and greater dopamine release correlates with greater increases in positive symptoms.

16.5 Provide evidence of the role of cortex in the negative and cognitive symptoms of schizophrenia.

Negative and cognitive symptoms are linked primarily to **hypofrontality** (decreased activity of the frontal lobes, particularly the dorsolateral prefrontal cortex or dlPFC). Evidence includes: 1. **Glutamate Hypothesis:** Decreased glutamate activity, such as from NMDA receptor antagonists like **PCP** and **ketamine**, produces hypofrontality and subsequently, negative and cognitive symptoms. 2. **Atypical Antipsychotics:** These drugs work by acting as **partial dopamine agonists** (e.g., aripiprazole), effectively **increasing dopamine activity where it is low** (dlPFC) while simultaneously **decreasing**

activity where it is high (mesolimbic pathway), thereby reducing both negative and positive symptoms.

16.6 Contrast the symptoms of major depressive disorder and bipolar disorder.

Major Depressive Disorder (MDD): Characterized by prolonged, unremitting periods of **depression**. **Bipolar Disorder:** Characterized by cyclical periods alternating between **mania** (extreme elation, energy, often not justified by external events) and **depression**. Both are characterized by disordered feelings.

16.7 Describe evidence that supports a genetic contribution to the development of affective disorders.

Hereditary factors contribute substantially to affective disorders. Evidence includes: 1. **Family History:** Close relatives of affected individuals are approximately 10 times more likely to develop an affective disorder. 2. **Twin Studies:** Concordance rates are high, reaching **69 percent in monozygotic twins** compared to 13 percent in dizygotic twins. 3. **Specific Genes:** Genes related to **circadian rhythms** (like *RORA* and *RORB*) and genes coding for glutamate receptors (like *GRM8*) are associated with MDD and bipolar disorder, respectively.

16.8 List the types of biological treatments for affective disorders.

Biological treatments include: 1. **Pharmacological Treatments:** **MAO inhibitors** (increase monoamines), **Tricyclic Antidepressants** (inhibit reuptake of 5-HT and NE), **SSRIs** (Selective Serotonin Reuptake Inhibitors, e.g., Prozac), and **SNRIs** (Serotonin and Norepinephrine Reuptake Inhibitors). **Ketamine** (NMDA antagonist) is used for treatment-resistant depression. **Lithium** is primarily used for bipolar disorder. 2. **Electroconvulsive Therapy (ECT):** Induces a seizure to alleviate severe symptoms. 3. **Neural Stimulation: Vagus Nerve Stimulation (VNS)** and **Transcranial Magnetic Stimulation (TMS)**. **Deep Brain Stimulation (DBS)** is used for treatment-resistant depression.

16.9 Describe the role of the frontal cortex in depression.

The frontal cortex plays a critical role in depression and mood regulation. Depressed patients consistently show **hyperactivity in the subgenual Anterior Cingulate Cortex (ACC)** and **decreased activity in other frontal**

16.10 Summarize the evidence in support of the monoamine theory of affective disorders.

regions (like the dorsolateral and ventromedial prefrontal cortex). Successful antidepressant treatments (including drugs, ECT, and DBS) reliably **decrease activity in the subgenual ACC** and typically increase activity in other prefrontal regions.

The **Monoamine Hypothesis** suggests that depression is caused by **insufficient activity of monoaminergic neurons** (specifically norepinephrine and serotonin).

Evidence includes: 1. **Pharmacology:** Drugs that elevate monoamine levels (like MAO inhibitors, SSRIs, and SNRIs) successfully treat depression, suggesting monoamine deficiency. 2. **Tryptophan Depletion:** The **tryptophan depletion procedure** reduces serotonin synthesis in the brain, causing depressive symptoms to reappear in treated patients, supporting serotonin's role in mood regulation.

16.11 Critique the evidence for the role of the promoter region of the 5-HT transporter gene in affective disorders.

The promoter region of the **5-HT transporter gene** has **short and long forms**. Some studies found that individuals with one or two **short alleles** who experienced stressful life events had a **higher probability of developing depression**. However, the overall role of this promoter region is currently **unclear** and needs further research, as some meta-analyses combining results have concluded that no significant association exists.

16.12 Explain the relationship between affective disorders, their treatment, and neurogenesis.

Stress and depression are associated with **reduced hippocampal neurogenesis** (the production of new neurons in the dentate gyrus). Conversely, **antidepressant treatments** (SSRIs, ECT, etc.) **increase hippocampal neurogenesis** in laboratory animals. The **therapeutic lag** (the time it takes for drugs to work, usually several weeks) is approximately the same as the time it takes for newly generated neurons to mature, suggesting neurogenesis may be involved in the drugs' delayed effect. Exercise is also correlated with increased

blood volume in the dentate gyrus in humans, suggesting it promotes neurogenesis.

16.13 Summarize the involvement of circadian rhythms in affective disorders.

Sleep disturbances are a major symptom of affective disorders, often involving reduced slow-wave sleep, fragmented sleep, and an early onset of REM sleep. **Sleep deprivation** can rapidly reduce depressive symptoms in some individuals, suggesting that sleep itself may be producing a "depressogenic" substance. **Seasonal Affective Disorder (SAD)**, a unique form of depression, is associated with short winter days. SAD can be effectively treated with **phototherapy** (exposure to bright light), which acts as a **zeitgeber** (time giver) to re-synchronize circadian rhythms regulated by the suprachiasmatic nucleus (SCN).

16.14 List the symptoms of anxiety disorders.

Anxiety disorders are the most common psychiatric disorders, characterized by **unrealistic, unfounded fear and anxiety**. They include specific diagnoses such as **panic disorder**, generalized anxiety disorder, and social anxiety disorder. Symptoms often include **tension, overactivity of the autonomic nervous system, expectation of disaster, and continuous vigilance for danger**.

16.15 Describe factors associated with the development of anxiety disorders.

Anxiety disorders have a **hereditary component**, often sharing genetic factors with affective disorders. Functional imaging studies show abnormal activity in brain regions associated with fear and emotion: 1. **Amygdala**: Shows **increased activation**. 2. **Ventromedial Prefrontal Cortex (vmPFC)**: Shows **decreased activation** (which normally inhibits fear/anxiety responses). Mutations in genes regulating the **BDNF** (brain-derived neurotrophic factor) protein are also associated with anxiety.

16.16 Summarize pharmacological treatments for anxiety disorders.

1. **Benzodiazepines:** Act as a **GABAA receptor agonist**. They are fast-acting, often used for **emergency treatment** (e.g., panic attacks), but are inappropriate for long-term use due to sedation, addiction risk, and tolerance. 2. **SSRIs:** Are **serotonergic agonists** and are the **preferred long-term treatment option**, typically used in conjunction with cognitive behavioral therapy.

16.17 List the symptoms of OCD.

Obsessive-Compulsive Disorder (OCD) involves **obsessions** (unwanted, persistent thoughts, images, or urges) and **compulsions** (uncontrollable, repetitive behaviors performed to reduce the distress from obsessions). Obsessions often involve germs, fear of danger, or need for exactness. Compulsions typically fall into four categories: **counting, checking, cleaning, and avoidance**. Individuals with OCD recognize that their thoughts and behaviors are irrational.

16.18 Describe the roles of genetic and environmental factors in the development of OCD.

Genetic Factors: Heredity plays a role, evidenced by a greater **concordance rate in monozygotic twins**. **Environmental Factors & Neural Changes:** OCD can follow **brain damage** (birth trauma, encephalitis, head trauma). Symptoms are associated with dysfunction or damage to the **basal ganglia**, and consistently show **increased activity in the frontal lobe and caudate nucleus**. Some cases are linked to a **Group A β -hemolytic streptococcal infection**, which triggers an autoimmune reaction damaging the basal ganglia.

16.19 Summarize treatments for OCD.

1. **Pharmacological Treatment: Serotonergic agonists** (e.g., clomipramine, fluoxetine) are effective, presumably by increasing activity in serotonergic inhibitory pathways. 2. **Surgical Interventions: Deep Brain Stimulation (DBS)** aimed at the basal ganglia (subthalamic nucleus) or tracts connected with it (e.g., internal capsule) can reduce symptoms in severe, treatment-resistant cases. **Cingulotomy** (surgical destruction of fiber bundles in the subcortical frontal lobe) is also a psychosurgical option.

Chapter 17: Neurodevelopmental Disorders, Stress, PTSD, and Substance Use Disorders

Learning Objective

Detailed Answer Key

17.1 Describe the symptoms and possible causes of autism spectrum disorder.

Symptoms: Require persistent deficits in **social communication and social interaction** (e.g., difficulty inferring others' intentions, avoiding eye contact) and **restricted, repetitive patterns of behavior** (e.g., stereotyped movements, insistence on routines, fixated interests). Symptoms are present from early childhood and are typically more common in males (4:1 ratio). **Possible Causes:** 1. **Genetic Factors:** Strongly hereditary (70–90% concordance in monozygotic twins) linked to **rare mutations** interfering with neural development. 2. **Neural Abnormalities:** Brain regions show unusual growth patterns, leading to excess **short-range axons** (local hyperconnectivity) and inhibiting **long-range axons**. Deficits are observed in the **superior temporal sulcus (STS)** and **medial prefrontal cortex** (regions for Theory of Mind) and abnormal **mirror neuron system** development. Lack of interest in faces leads to failure of the **fusiform face area (FFA)** to activate normally. 3. **Hormonal/Peptide Changes:** Elevated prenatal **testosterone** levels and lower levels of **oxytocin** are correlated with symptoms.

17.2 Describe the symptoms and possible causes of ADHD.

Symptoms: Require a persistent pattern of **inattention** (e.g., difficulty sustaining attention, easily distracted) and/or **hyperactivity and impulsivity** (e.g., inability to sit still, acting without reflection) that interferes with learning, persisting for at least six months. It is the most common childhood behavior disorder and is associated with aggression, anxiety, and learning disabilities. **Possible Causes:** 1. **Heredity:** The disorder is highly heritable (75–91% concordance). 2. **Neural Dysfunction:** Symptoms resemble those caused by damage to the **prefrontal cortex**. The prevailing hypothesis is that ADHD is caused by **underactivity of dopaminergic transmission**

(hypodopaminergic activity) in the prefrontal cortex. The effects of dopamine follow an **inverted U-shaped curve**.

Both systems initiate the stress response (fight-or-flight): 1.

SAM System (Sympathetic Adrenal-Medullary System):

The **hypothalamus** $\rightarrow$ **sympathetic nervous system** $\rightarrow$ **adrenal medulla**. It releases **epinephrine and norepinephrine (catecholamines)**, initiating a **rapid activation** of the sympathetic nervous system, increasing heart rate and energy supply for immediate action. 2. **HPA Axis (Hypothalamic Pituitary Adrenal Axis):** The **PVN of hypothalamus** $\rightarrow$ **anterior pituitary (ACTH)** $\rightarrow$ **adrenal cortex**. It secretes **glucocorticoids (like cortisol)**, which mobilize protein and fat stores to increase glucose availability for energy. This is a slower, hormonal response.

17.3 Compare the SAM system and HPA axis in coordinating a stress response.

The harmful effects of long-term stress are primarily caused by **prolonged secretion of glucocorticoids**. These effects include: **increased blood pressure** (leading to cardiovascular disease), damage to muscle tissue, **suppression of the immune system** (leading to slower wound healing and increased susceptibility to infectious illness), steroid diabetes, and infertility. The chronic "wear and tear" on the body is referred to as **allostatic load**.

17.4 Describe the negative health outcomes associated with chronic stress.

Long-Term Glucocorticoid Exposure: Prolonged exposure destroys neurons in the **hippocampal formation** (specifically the CA1 field), increasing the likelihood of memory problems with age. Intense, long-term stress can even cause atrophy of neurons in the dorsomedial prefrontal cortex. **Early Nurturing Experiences (Resilience):** Early life nurturing experiences (e.g., increased maternal licking/grooming in animals, following a stressor) reduce the adult organism's reactivity to stress by attenuating the **HPA axis response** (reducing CRH production).

17.5 Compare the effects of long-term glucocorticoid exposure and early nurturing experiences on the brain in response to stress.

17.6 Summarize the relationship between stress and infectious disease.

The stress response impairs the functions of the **immune system**. Studies show that people experiencing **stressful situations** (e.g., job loss, chronic caregiving, hostile interactions) are more susceptible to illness, such as upper respiratory infections and the common cold. This is because prolonged stress hormones (glucocorticoids) can suppress the immune system.

17.7 List the symptoms of PTSD.

PTSD symptoms follow exposure to a traumatic event (especially violence from others). Symptoms include: 1. **Re-experiencing:** Recurrent **dreams, recollections, or "flashbacks"** of the traumatic event. 2. **Avoidance/Detachment:** Avoiding stimuli related to the event, diminished interest in social activities, feelings of **detachment**. 3. **Hyperarousal/Negative Changes:** **Difficulty sleeping**, irritability, **outbursts of anger**, heightened startle response, and impaired mental health.

17.8 Describe the roles of genetic and environmental factors in the development of PTSD.

Environmental Factors: Risk increases with the **number of traumatic events** experienced. **Genetic Factors:** Genetic factors contribute to susceptibility, possibly by sharing genes with anxiety and depressive disorders. PTSD is associated with a **smaller hippocampus**. Twin studies suggest that a smaller hippocampus may be a **preexisting genetic vulnerability** to PTSD, rather than stress-induced damage. **Neural Mechanisms:** PTSD involves increased activity in the **amygdala** (emotional reactions) and **smaller activation of the prefrontal cortex** (impaired inhibition/extinction of fear).

17.9 Summarize treatments for PTSD.

Treatments include **Cognitive Behavior Therapy (CBT)** and **antidepressant (SSRI) medication**. **TMS** (transcranial magnetic stimulation) of the dorsolateral prefrontal cortex has also shown efficacy in reducing symptoms. **Preventative Treatment:** Immediately following trauma, administering **cortisol** (to increase negative feedback to the HPA axis) or a **beta-adrenergic antagonist** (like

propranolol, to suppress emotional memory consolidation) may reduce the subsequent development of PTSD.

17.10 Describe how positive and negative reinforcement contribute to the development of substance use disorders.

Positive Reinforcement: The immediate **pleasurable effect** (e.g., euphoria) of the drug acts as an appetitive stimulus that immediately reinforces the preceding drug-taking behavior. Effectiveness is maximized when the reinforcing stimulus immediately follows the response (e.g., rapid onset of heroin high vs. slow onset of methadone). **Negative Reinforcement:** Drug use is reinforced because the behavior **removes or reduces an aversive state**, such as anxiety or the unpleasant symptoms of withdrawal.

17.11 Describe commonly abused drugs.

All commonly abused drugs trigger the release of **dopamine in the nucleus accumbens (NAC)**. 1. **Opiates (e.g., Heroin, Fentanyl):** Stimulate opiate receptors, producing analgesia and reinforcement (via VTA/NAC pathway). Rapid crossing of the blood-brain barrier contributes to high abuse potential. 2. **Stimulants (e.g., Cocaine, Amphetamine):** Potent dopamine **agonists**; Cocaine blocks dopamine reuptake, while amphetamine stimulates dopamine release (causing transporters to run in reverse). 3. **Nicotine:** Stimulates **nicotinic acetylcholine receptors** in the VTA, causing dopamine release in the NAC. Accounts for more deaths than other drugs. 4. **Alcohol:** Acts as an indirect **NMDA receptor antagonist** and an indirect **GABAA receptor agonist**. Reinforcement is partially mediated by stimulating the release of **endogenous opioids**. 5. **Cannabis (THC):** Stimulates the **CB1 receptor**, producing euphoria and memory impairment (by disrupting hippocampal function).

17.12 Summarize the research on effective treatment for substance use disorders.

Successful treatment is challenging, with abstinence rates of 40-60% after a year. 1. **Opiates: Methadone maintenance** (slow-acting agonist that prevents high) and **buprenorphine** (partial opiate agonist that blocks effects but has weak abuse potential). 2. **Stimulants (Cocaine): Vaccines** (to produce antibodies that block the drug from

entering the brain) are under investigation. 3. **Nicotine:**
Nicotine maintenance therapy (patches/gum); **bupropion**
(antidepressant, catecholamine reuptake inhibitor);
varenicline (nicotinic partial agonist). 4. **Alcohol:**
Naltrexone (μ opiate antagonist, reduces reinforcing
value) and **acamprosate** (NMDA receptor antagonist).